



# basic education

Department:  
Basic Education  
**REPUBLIC OF SOUTH AFRICA**

## **SENIOR CERTIFICATE EXAMINATIONS/ NATIONAL SENIOR CERTIFICATE EXAMINATIONS**

**ENGINEERING GRAPHICS AND DESIGN P2**

**MAY/JUNE 2024**

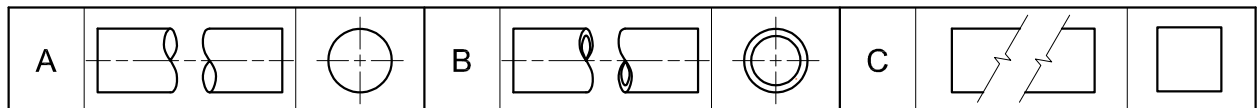
**MARKING GUIDELINES**

**MARKS: 100**

**These marking guidelines consist of 7 pages.**

| ANSWERS |                                      |    |
|---------|--------------------------------------|----|
| 1       | SIPHO                                | 1  |
| 2       | www.mecdrow.co.za                    | 1  |
| 3       | JJVWBJ-6                             | 1  |
| 4       | 1500                                 | 1  |
| 5       | CAD                                  | 1  |
| 6       | 2                                    | 1  |
| 7       | 18                                   | 1  |
| 8       | 1 / ONE                              | 1  |
| 9       | 4                                    | 1  |
| 10      | ROUNDING                             | 1  |
| 11      | C: 39 D: 28 E: 106 F: Ø38 G: 90      | 5  |
| 12      | 35°, 36°, 37°                        | 1  |
| 13      | M14 x 1,5                            | 2  |
| 14      | MAX. = 162,02 MIN. = 161,83          | 2  |
| 15      | 9                                    | 1  |
| 16      | Q: T                                 | 2  |
|         | R: V                                 |    |
| 17      | AUXILIARY VIEW                       | 1  |
| 18      | DIRECTION OF LAY/ CIRCULAR MACHINING | 1  |
| 19      | MATERIAL REMOVAL NOT PERMITTED       | 1  |
| 20      | See below                            | 4  |
| TOTAL   |                                      | 30 |

## ANSWER 20



FREEHAND ①

CIRCLE ①

2 x CL ①

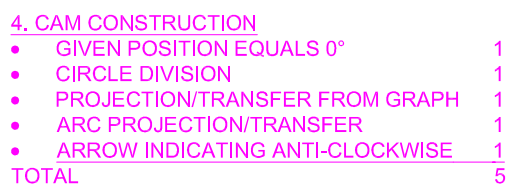
OUTER CIRCLE ①

INNER CIRCLE ①

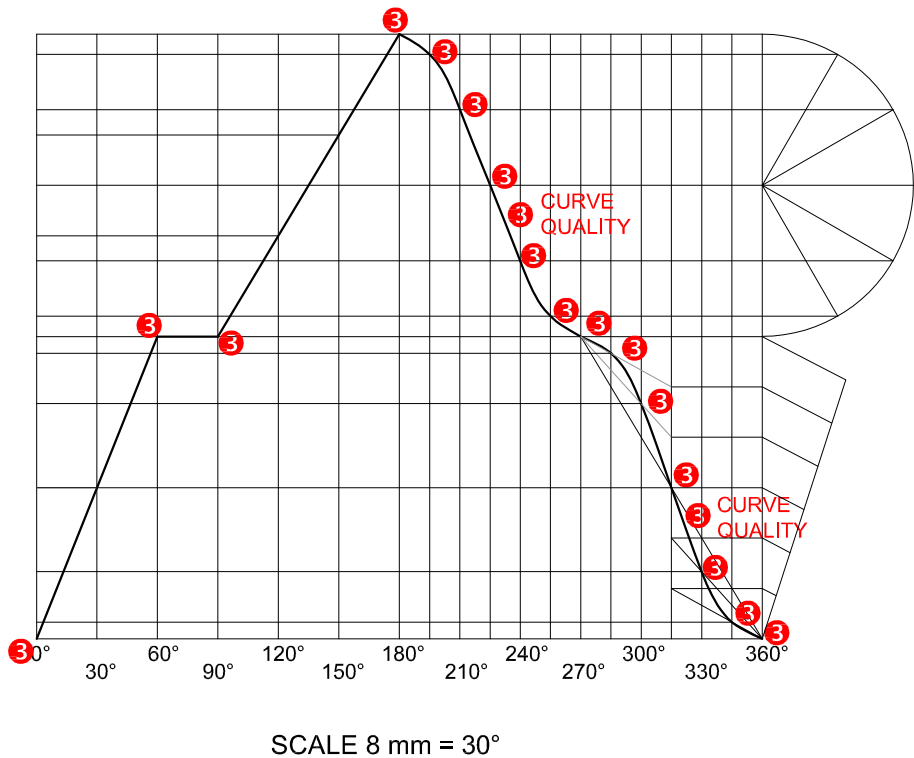
2 x CL ①

SQUARE / RECTANGLE ✓<sub>1</sub>

PAPER 2 QUESTION 1  
 GRADE 12  
 DBE MAY/JUNE 2024  
 MARKING GUIDELINES

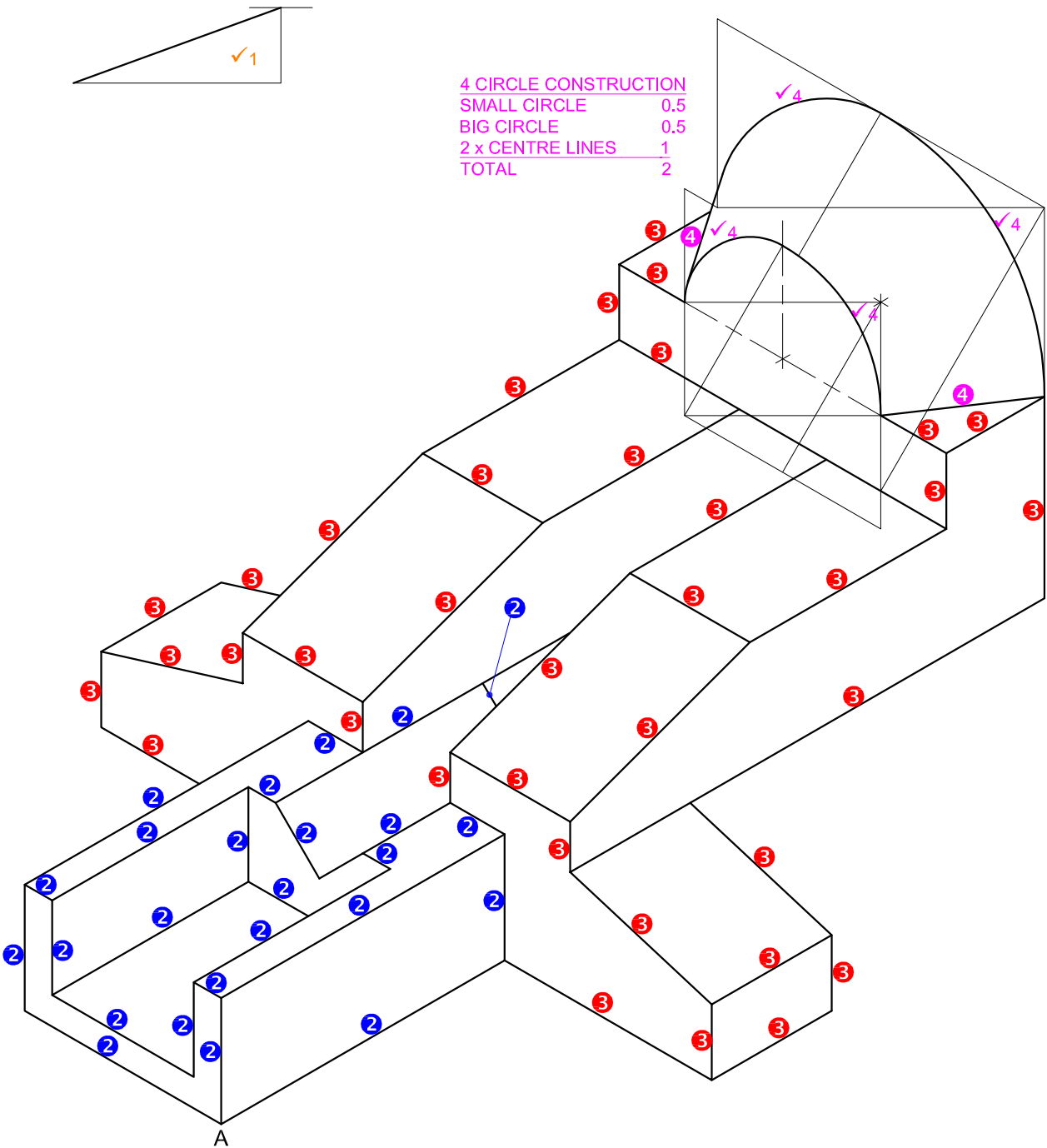


PAPER 2 QUESTION 2  
GRADE 12  
DBE MAY/JUNE 2024  
MARKING GUIDELINES



|                                           |  |     |
|-------------------------------------------|--|-----|
| <u>2. GRAPH CONSTRUCTION</u>              |  |     |
| • BASE LENGTH 96 mm                       |  | 1   |
| • 12 EQUAL DIVISIONS                      |  | 1   |
| • TOTAL HEIGHT 80 mm                      |  | 0.5 |
| <u>SIMPLE HARMONIC MOTION</u>             |  |     |
| • SEMI CIRCLE                             |  | 0.5 |
| • CIRCLE DIVISIONS                        |  | 0.5 |
| • 15° SUBDIVISIONS                        |  | 0.5 |
| • PROJECTION                              |  | 0.5 |
| <u>UNIFORM ACCELERATION + RETARDATION</u> |  |     |
| • LINE DIVISION                           |  | 0.5 |
| • 15° SUBDIVISIONS                        |  | 0.5 |
| • FIRST $\frac{1}{2}$ CONSTRUCTION        |  | 0.5 |
| • SECOND $\frac{1}{2}$ CONSTRUCTION       |  | 0.5 |
| <u>LABEL</u>                              |  |     |
| TOTAL                                     |  | 7   |

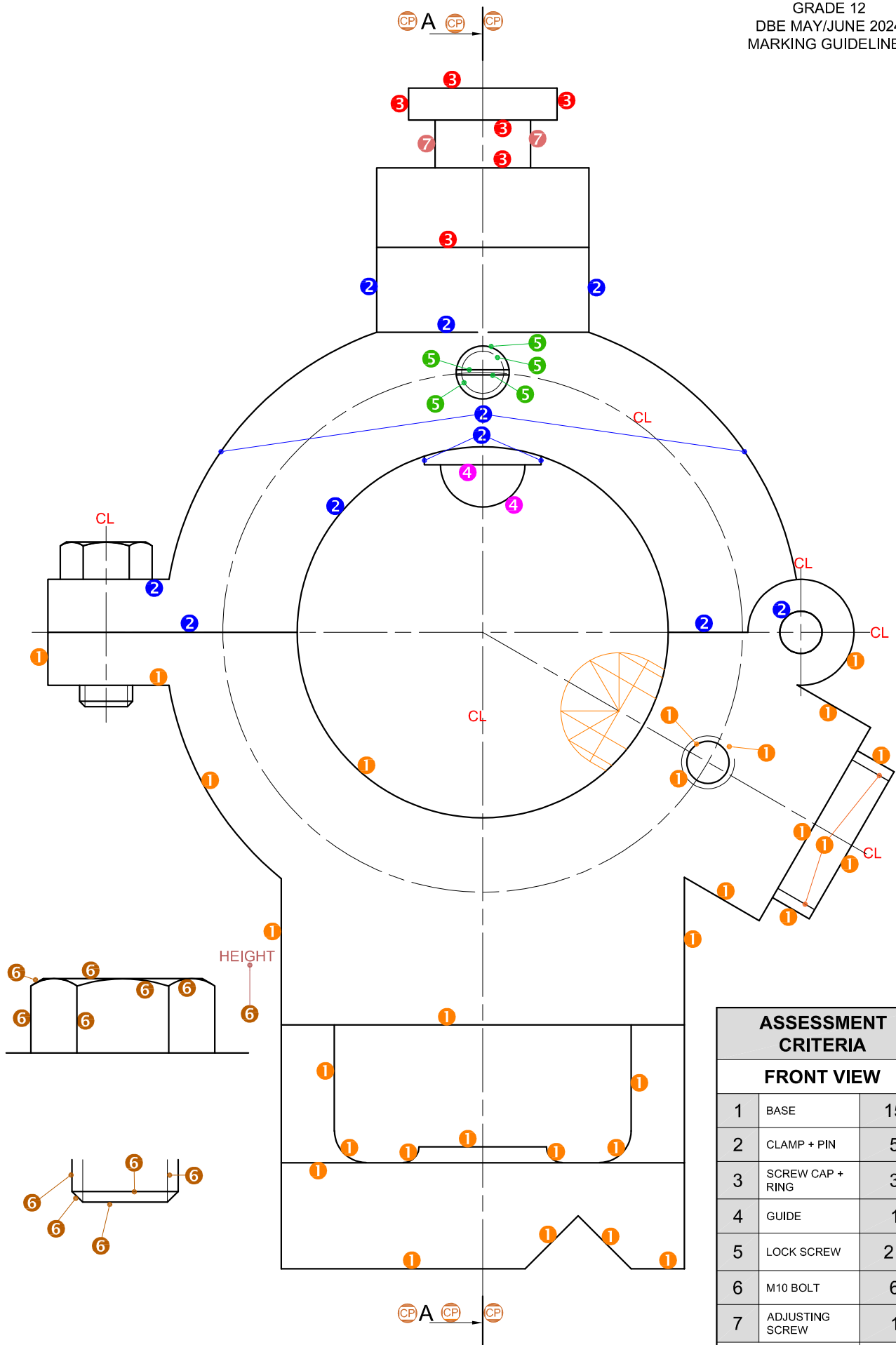
| ASSESSMENT CRITERIA |                                  |    |
|---------------------|----------------------------------|----|
| 2                   | GRAPH CONSTRUCTION               | 7  |
| 3                   | PLOTTING GRAPH +<br>GRAPH CURVES | 9  |
| TOTAL               |                                  | 16 |

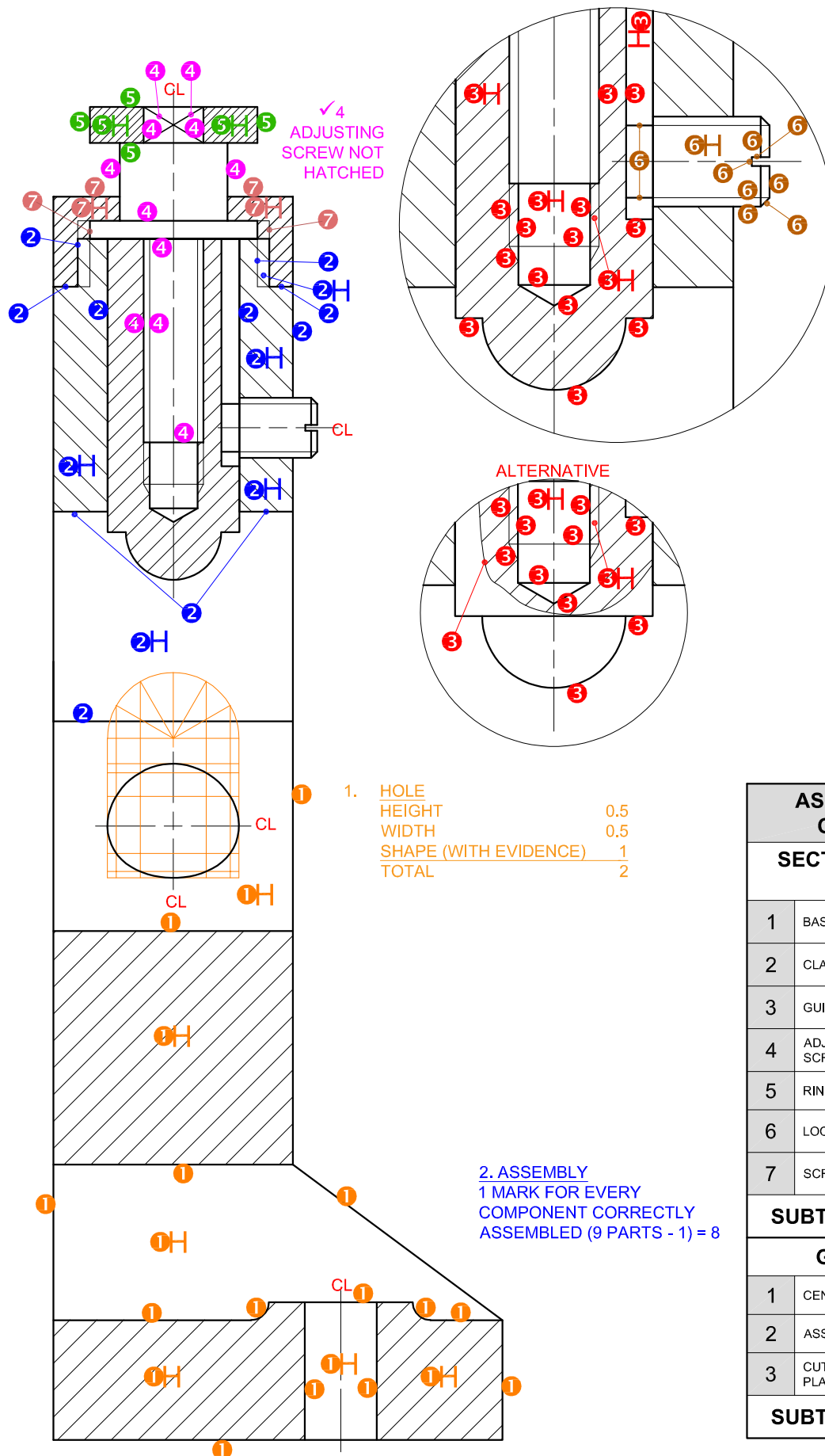


✓1 ORIENTATION / PLACEMENT

| ASSESSMENT CRITERIA |                     |                                |
|---------------------|---------------------|--------------------------------|
| 1                   | PLACING + AUX. VIEW | 2                              |
| 2                   | FRONT PORTION       | 12 <sup>1</sup> / <sub>2</sub> |
| 3                   | UPPER PORTION       | 18 <sup>1</sup> / <sub>2</sub> |
| 4                   | ARCS + CONSTR. + CL | 7                              |
| TOTAL               |                     | 40                             |

PAPER 2 QUESTION 4  
GRADE 12  
DBE MAY/JUNE 2024  
MARKING GUIDELINES





✓ 4  
ADJUSTING  
SCREW NOT  
HATCHED

ALTERNATIVE

1. HOLE  
HEIGHT 0.5  
WIDTH 0.5  
SHAPE (WITH EVIDENCE) 1  
TOTAL 2

2. ASSEMBLY  
1 MARK FOR EVERY  
COMPONENT CORRECTLY  
ASSEMBLED (9 PARTS - 1) = 8

1. CENTRE LINE MARK ALLOCATION  
SANS COMPLIANT TYPE 1  
CENTRE LINES (1 - 3 = 0,5; 4 - 7 = 1; 8 & 9 = 2; 10 & 11 = 3) 3  
4

### ASSESSMENT CRITERIA

#### SECTIONAL LEFT VIEW

|                 |                    |                                    |
|-----------------|--------------------|------------------------------------|
| 1               | BASE               | 12                                 |
| 2               | CLAMP              | 7                                  |
| 3               | GUIDE              | 8 $\frac{1}{2}$                    |
| 4               | ADJUSTING<br>SCREW | 6 $\frac{1}{2}$                    |
| 5               | RING               | 3                                  |
| 6               | LOCK SCREW         | 4 $\frac{1}{2}$                    |
| 7               | SCREW CAP          | 3                                  |
| <b>SUBTOTAL</b> |                    | <b>44 <math>\frac{1}{2}</math></b> |

#### GENERAL

|                 |                      |           |
|-----------------|----------------------|-----------|
| 1               | CENTRE LINES         | 4         |
| 2               | ASSEMBLY             | 8         |
| 3               | CUTTING<br>PLANE A-A | 3         |
| <b>SUBTOTAL</b> |                      | <b>15</b> |

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